Humic Cambisols

Description
Humic Cambisols are moderately developed mineral soils characterized by a dark, humus-rich surface horizon over a weakly developed subsurface (cambic) horizon that shows initial alteration of parent material.
They are called **“Humic”** because of their **high organic matter content** in the upper horizon (usually >1.5–2.5% organic carbon), which gives the surface layer a dark color and improves structure, nutrient retention, and biological activity.
These soils are common in **humid tropical and temperate regions**, typically forming on **volcanic ash, basic igneous rocks, or well-weathered parent materials** in **moderately sloping uplands**. They represent a transition stage between young soils (like Regosols) and more developed soils (like Luvisols or Nitisols).

Drainage:

- **Wet season:** Generally well-drained due to good soil structure and moderate porosity; may experience short-term saturation in depressions.
- **Dry season:** Retains moderate moisture thanks to humus content; does not harden excessively.
- **Landscape:** Common on gentle to moderately sloping uplands, lower mountain slopes, and volcanic footslopes; stable, erosion-resistant surfaces with active biological mixing.

Depth:

- Moderately deep to deep (**80–150 cm**).
- Dark humic horizon usually 20–40 cm thick, underlain by weakly developed cambic (Bw) horizon showing initial clay formation and structure.
- Rooting depth is generally unrestricted, promoting good root penetration and nutrient uptake

Workability:

- **When wet:** Friable and plastic but not sticky; good tilth and easy cultivation.
- **When dry:** Remains crumbly and workable; structure resists crusting and compaction.
- Excellent aggregate stability allows both manual and mechanized farming.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Moderate to high (**15–30 cmol(+)/kg**) due to humus and weathered clay minerals.
- **Base saturation:** Moderate to high (**40–80%**) depending on parent material and rainfall regime.
- **Organic matter:** High in surface horizon, enhancing nutrient retention and microbial activity.

Nutrients:

- Nitrogen and phosphorus may decline under continuous cropping without replenishment.
- Potassium and calcium are often sufficient in volcanic or basic rock-derived soils.
- Micronutrients (Zn, Mn, Fe, Cu) are generally adequate.
- Excellent buffering capacity due to the humus-clay complex.

pH:

- Slightly acidic to near neutral (**5.5–6.8**).
- Can be more acidic in high-rainfall regions or where leaching is intense.

Management:

- Maintain soil organic matter through compost, green manure, and crop residue incorporation.
- Apply balanced N and P fertilizers to sustain productivity, particularly under intensive cropping.
- Use contour farming or terracing on sloping terrain to reduce erosion.
- Avoid over-tillage to preserve structure and humus.
- Suitable for **maize, beans, potatoes, coffee, tea, bananas, and vegetables** due to stable structure and moderate fertility.